# Quantifying the Impact of Temporal and Causal Structures on Narrative Similarity Retrieval

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## Abstract

This research proposal investigates whether explicitly modeling temporal and causal relations between events improves narrative similarity retrieval beyond event-only and implicit dense embedding approaches. While current state-of-the-art models achieve strong retrieval performance, they often rely on surface-level lexical cues or sacrifice structural interpretability. The project proposes a novel structured narrative representation combined with a controlled ablation framework integrating supervised event extraction (via MAVEN dataset) and LLM-based temporal and causal inference. Structured event graphs will be encoded using multiple instruction-tuned embedding architectures and evaluated on the Tell Me Again! dataset via information retrieval metrics. The study quantifies the independent and joint contributions of temporo-causal structure across embedding backbones.

## 1 Introduction

Computational narrative analysis has gained increasing attention in recent years, with applications across economics, politics, and psychology [5]. At its core, narrative concerns what occurs in a story rather than how it is told [9]. Methodologically, the field has evolved from keyword-based approaches to Large Language Models (LLMs) capable of bridging linguistic features and theoretical narrative constructs [20]. This shift is particularly valuable for identifying narrative similarity at scale, enabling retrieval and QA systems that can detect re-tellings or structurally similar stories despite surface-level differences [3, 8].

However, the domain faces two core challenges. The first is to distinguish surface-level lexical semantics from deep narrative structures. Standard models often over-rely on vocabulary or named entities (e.g., location names) rather than inherent plot logic [9]. Then, models cannot find similarity between similar versions of stories that belong to same event, since they can diverge within their vocabulary and style drastically.

The second challenge is modeling the dynamic logic of the narratives. Here, identifying *what* happens (i.e., extracting events) can be insufficient without understanding *when* (temporality) and *why* (causality) they occur [17, 30]. Although extracting events from narratives is relatively a mature field where state-of-the-art (SoTA) models reach high performance on established benchmarks, extracting the temporal or causal relationships is challenging because it requires "common sense reasoning" to interpret these relationships within narratives [17], which makes the development of hand-built scripts time-consuming [17, 30, 31].

Previous work has modeled narratives as bags of events and character roles, extracted character-centered structures using LLMs, or learned dense summary embeddings [3, 5, 8, 20, 28]. However, existing approaches either omit temporal and causal structure, lack interpretability, or exhibit entity bias (i.e., learning specific names rather than underlying relationships) [5, 7, 9]. Furthermore, [7] demonstrated that story embeddings trained in local event chain prediction (Narrative Cloze task) fail to identify similar stories, suggesting that more complex structural representations of narratives are needed. Thus, there is a lack of studies in the literature focused on how explicit temporal and causal annotations impact event representations when using narrative similarity as a proxy of success.

To our knowledge, no study has systematically quantified the incremental contribution of temporal and causal modeling over event-only representations for narrative similarity. Addressing this gap enables a clearer assessment of whether structured, interpretable representations provide measurable gains beyond opaque LLM-based embeddings. We therefore combine supervised event extraction with LLM-based temporal and causal "commonsense" inference to evaluate their joint contribution.

Methodologically, we distinguish between the detection of events and detection of relationships by employing a SoTA supervised event extractor to robustly identify narrative events, followed by leveraging zero-shot reasoning of LLMs to infer the temporal and causal links. Then, the research (sub)question(s) are as follows: *To what extent does the explicit modeling of the narratives through events and LLM-inferred temporal and causal relations improve the retrieval of narratively similar stories, compared to implicit dense vector representations?*

- Does event-based narrative representation outperform verb-based lexical baselines in narrative similarity retrieval?
- What are the independent and joint effects of adding explicit temporal and causal relations to event-based representations?
- To what extent jointly modeled temporo-causal structures provide additional gains over separately integrated temporal and causal augmentations?
- Are the retrieval gains observed when moving from event-only representations to temporally enriched, causally enriched, and jointly temporo-causal representations consistent across different embedding architectures (Sentence-BERT and E5)?
- How does the best-performing explicit structural model compare to strong implicit dense narrative model (StoryEmbed) and traditional lexical methods (BoW and TF-IDF)?

## 2 Related Work

Despite advances in narrative modeling, current state-of-the-art approaches rely either on surface-level lexical overlap or implicit dense embeddings, which suffer from entity bias and limited interpretability [3, 5, 8, 20, 28]. Although temporal and causal relation extraction has been extensively studied [15, 16, 25], their independent and joint contributions to narrative similarity retrieval have not been systematically evaluated.

### 2.1 Narrative Representation and Similarity

Modeling narratives through events and structural features has long been central to computational narratology. Prior work has represented narratives as event chains [2], plot structures [6], character roles [24], emotional arcs [23], and turning points [19]. Building on Genette's theory, [20] which emphasize narrative's multidimensional nature across time, setting, and perspective. However, despite offering rich structural abstractions, these approaches do not provide scalable methods for mechanically comparing narratives [3].

A computational approach to framing narrative similarity as an information retrieval task was introduced by [3], who proposed a benchmark of movie plot summaries. They represent stories as unordered sets of event triggers and character roles. This abstraction improved retrieval over a simple baseline approach, namely bag-of-words (BoW), but not substantially (8%). The authors attributed this to information loss from structural abstraction and the absence of temporal or sentiment modeling. Subsequent work, StoryEmbed, sought to reduce information loss due to structural abstractions: [8] fine-tunes LLMs to produce summary-level embeddings for retrieving retellings of the same story. Although achieving strong performance ( 85% P@N), results drop substantially ( 65% P@N) when character names are anonymized. This drop suggests that their model relied on surface-level lexical cues, namely entity bias, rather than deeper narrative logic.

More recently, [5] showed that LLMs can extract structured narrative components and represent stories through role-based character embeddings. While effective for clustering similar news articles, the approach remains primarily character-centric and typically relies on a single extracted actor per role. As a result, narratives with similar role configurations but different temporal progression or causal structure may appear structurally equivalent.

### 2.2 Event-based Narrative and Event Extraction

One of the commonality between all SoTA methods for event detection is the reliance on pretrained language models as their backbone [12]. Transformer-based architectures such as BERT and its variants dominate contemporary event detection pipelines, yet no single architecture has emerged as consistently superior across benchmarks. Rather, performance differences often depend on how events are theoretically defined and operationalized within specific ontologies [12]. Furthermore, transformer-based semantic event extraction (T-SEE) has been shown to outperform LLM-based prompting (L-SEE) methods in structured event classification, particularly when the ontologies are larger and more fine-grained [11].

Regarding the training resources of these models, large-scale benchmark datasets such as Automatic Content Extraction (ACE) and a MAssive eVENt detection dataset (MAVEN) provide fine-grained event ontologies that are beyond simple event triggers (verbs), including abstract event types (e.g., hurricane, war) across diverse domains [27, 30]. Unlike ACE, MAVEN offers a publicly available alternative without licensing restrictions; and also, supports large-scale supervised training. In particular, MAVEN enables supervised transformer models to learn robust mappings between lexical cues and structured event classes, which results in relatively high performance in event trigger identification. In addition, MAVEN provides standardized negative trigger annotations, which reduces variability in negative sampling strategies and helps with consistent training and evaluation setups.

Importantly, while event detection has reached a relatively stable level of performance, the annotation of semantic links between events remains substantially more challenging. As [17] observe, the temporal and causal annotation between events requires greater interpretive effort and yields only moderate agreement between annotators. Thus, unlike event detection, inter-event relation modeling, which involves discourse-level reasoning, motivates us to the use of LLM-based inference in our framework.

### 2.3 Temporal and Causal Structure in Narratives

Communicating events is in the center of human languages, and in order to understand them, it is necessary to uncover the complex connections between these events [29]. Event relation extraction (ERE) – including temporal, causal, and coreference relations – has therefore emerged an important challenge in NLP [13]. Furthermore, modeling inter-event relations supports downstream applications such as story generation, common sense reasoning, question answering, and narrative forecasting [29]. Chiefly, narrative coherence depends not only on which events occur, but on how they are temporally ordered and causally connected.

*2.3.1 Modeling Temporal Relations.* Temporal relation extraction has been extensively studied within the TimeML framework [21], which operationalizes temporal relations between events and temporal expressions through TLINK annotations. TimeML operationalizes temporal reasoning using relations derived from Allen's interval algebra, which creates a partial order of events within a text. Subsequent shared tasks, such as TempEval, further standardize temporal relation modeling as a pairwise classification problem over annotated events [26]. More recent annotation efforts, such as the THYME guidelines, simplify the temporal inventory to a smaller set of core relations (e.g. BEFORE, OVERLAP, CONTAINS) [18]. These approaches demonstrate that temporal structure is central to organizing event sequences, yet temporal relations are often annotated locally or constrained by sentence boundaries, which limits their capacity to represent broader narrative relationships [18].

*2.3.2 Modeling Causal Relations.* Unlike temporal relations, causal relation modeling remains less standardized. [15] explicitly note that while temporal modeling has achieved consensus through TimeML, annotation of causal relations lacks a unified framework. They introduce CLINK annotations inspired by TimeML to capture directed causal relations between events (e.g. CAUSE, ENABLE, PREVENT) categories based on theoretical foundations of causality.

However, causality often depends on common sense reasoning and contextual inference rather than purely lexical cues [17]. Additionally, causal relations impose temporal constraints: a cause must temporally precede or overlap with its effect [17].

*2.3.3 Joint Modeling of Temporal and Causal Relations.* Although temporal and causal relations are often studied independently, they are inherently intertwined in the narrative structure. The CaTeRS [17] framework is based on the assumption that script learning and story understanding require capturing both dimensions simultaneously. Hence, they created a unified annotation scheme for tempero-causal relations between events. A central insight of CaTeRS is that causal relations carry temporal implications: if event A causes event B, then A must begin before (or at least overlap with) B. For example, in the sentence "The explosion caused him to die," the causal relationship between the explosion and death implies a temporal ordering constraint. Yet, the two events may overlap in duration. CaTeRS therefore encodes relations such as cause-before or cause-overlap, enable, prevent, and cause-to-end, to capture both explanatory dependency and temporal structure within a single semantic link.

## 2.4 Narrative Embeddings

*2.4.1 Lexical and Early Embedding Baselines.* Before widespread adoption of contextual encoders, semantic similarity was commonly modeled using lexical vector space representations such as Bag-of-Words (BoW) [3] and TF-IDF [10]. These approaches represent texts as sparse vectors based on word frequency statistics and measure similarity through cosine distance. While computationally efficient and interpretable, they are heavily dependent on surface-level lexical overlap and are particularly sensitive to named entities and stylistic variation [4]. Early dense embedding approaches such as Doc2Vec extended this paradigm by generating fixed-length document embeddings without contextual token interaction, but lack the ability to model fine-grained semantic dependencies across tokens [8].

*2.4.2 LLMs as Encoders.* Recently, instruction-tuned embedding models such as E5 have emerged as strong backbones for semantic retrieval [5, 8, 28]. [28] demonstrate that E5 achieves SoTA performance on large-scale embedding benchmarks by fine-tuning decoder-based language models with contrastive objectives and diverse synthetic retrieval tasks. Unlike earlier BERT-style encoders trained primarily on natural language inference or limited labeled datasets, E5 is explicitly optimized for retrieval scenarios, aligning semantically related texts in embedding space. Although alternative adaptations of decoder-only architectures have been proposed to mitigate the limitations of causal attention when extracting embeddings, empirical evaluations suggest that such modifications do not produce substantial inference improvements over extensively fine-tuned E5 models [28].

In narrative applications, both [8] and [5] adopt E5 due to its empirical advantages over Sentence-BERT and related baselines in story-level similarity tasks. StoryEmbed, built on an E5-Mistral backbone, outperforms Sentence-T5 (XXL) and Doc2Vec on benchmarks such as Movie Remakes Dataset, yielding stronger retrieval performance than sentence-level encoders [8]. Although Doc2Vec sometimes benefits from reduced entity sensitivity under pseudonymization, E5-based models achieve superior overall results. Similarly, [5] show that E5-large outperforms MiniLM and BERTopic (typically Sentence-BERT-based), particularly in distinguishing structurally divergent narratives with similar topical content. Together, these results indicate that E5 provides a robust, retrieval-optimized embedding foundation for evaluating explicitly structured narrative representations against static and sentence-level baselines.

Although event extraction, temporal–causal modeling, and dense embeddings have advanced substantially, they have rarely been integrated within a unified narrative similarity framework. It remains unclear whether explicitly modeling event, temporal, and causal structure provides measurable improvements over event-only or fully implicit representations, while not sacrificing interpretability.

# 3 Methodology

## 3.1 Data

*3.1.1 Event extraction.* Event triggers are extracted using a supervised transformer-based event detection model trained on the MAVEN dataset [30], which provides a large-scale, fine-grained event ontology covering over 160 event types, including verbal, nominal, and abstract triggers in diverse domains. Furthermore, MAVEN enables the mapping of lexical variants (e.g., "attack", "assault", "bombing") to shared event types. We retain both the trigger span and its event type, as the type reduces lexical variability and strengthens structural narrative representation.

*3.1.2 Narrative Similarity.* We evaluate narrative similarity using the Tell Me Again! dataset [9], which contains approximately 96,831 independently written summaries of 29,505 underlying works, collected from Wikipedia. Summaries sharing the same work identifier form relevance clusters, which enables binary retrieval evaluation: summaries of the same work are treated as relevant, others as non-relevant. Because summaries vary in length and detail—from brief premises to full plot retellings, retrieval performance can be influenced by verbosity and abstraction. By explicitly modeling temporal and causal structure, our approach aims to capture core narrative logic beyond such surface variation.

*3.1.3 Dataset Scaling and Computational Efficiency.* Processing the entirety of Tell Me Again! dataset via LLMs and computing pairwise cosine similarities is computationally heavy. To manage costs, we conduct an exploratory data analysis (EDA) to construct a representative subset of the corpus. Based on the empirical distributions observed in the complete data set, we may apply stratified sampling to preserve the relative distributions of summary lengths and cluster sizes to maintain structural and narrative variance of the original corpus.

## 3.2 Extracting Events and Semantic Links

*3.2.1 Event Extraction Backbone.* The event triggers and types are extracted using a supervised transformer-based event detection model trained on the MAVEN dataset [30]. Among reported architectures, we adopt the Bi-directional Encoder Representations From Transformers with a conditional random field (BERT+CRF) configuration, which achieves strong performance on MAVEN and demonstrates robustness in sentences containing multiple events.

[30] show that BERT+CRF outperforms alternative architectures even without dynamic multi-pooling, attributed this advantage to the ability of the CRF layer to model correlations between multiple event triggers within the same sentence.

We deliberately employ a supervised transformer event-annotation backbone rather than a prompt-based LLM for event detection, in light of the latest comparison studies examined by [11]. Since event triggers serve as the structural backbone for subsequent temporal and causal modeling, we prioritize deterministic closed-set classification over generative flexibility.

However, it is important to acknowledge a potential domain shift: MAVEN is derived from encyclopedic Wikipedia articles on real-world events, whereas Tell Me Again! contains fictional plot summaries that range from brief premises to full retellings [9, 30]. This difference may lead to misclassified narrative-specific events. However, MAVEN's broad ontology (168 event types) and high density of overlapping event mentions make it a strong proxy for modeling the complex event structures found in plot summaries.

For each detected event, we retain both the trigger span and its corresponding event type:

$$e_i = (\text{trigger span}, \text{event type})$$

*3.2.2 Temporal and Causal Link Annotation.* Following event extraction, we annotate inter-event semantic relations with respect to temporality and causality. Earlier work relied on manual annotation of temporal and causal links; for example, [17] report that semantic link annotation requires substantial effort (around 11 minutes per story) and yields only moderate inter-annotator agreement [17]. To enable large-scale structural annotation, we employ a Large Language Model for relation inference.

We use Llama-3-8B-Instruct to annotate temporal and causal relations between extracted events. This open-source model achieves state-of-the-art zero-shot performance comparable to proprietary systems such as GPT-3.5 while maintaining strong narrative understanding and computational efficiency [5]. Its open weights enable full reproducibility through transparent parameters and deterministic inference, unlike commercial LLMs. Recent work further shows that Llama-3-8B can reliably operationalize narratological constructs into structured outputs, achieving performance comparable to GPT-4 in narrative analysis tasks [20].

To mitigate the inherent stochasticity of LLMs and ensure stability, we run model inference without sampling. By forcing the model to select the highest probability token at each step, the extraction process becomes reproducible and deterministic[5]. Furthermore, to control the output format and prevent hallucinated or inconsistently formatted relations, the LLM is prompted to return its predictions strictly as a structured JSON object. This is a constraint that has proven highly effective for extracting narrative roles [5]. To operationalize this extraction, we utilize a zero-shot prompting strategy guided by a structured codebook. Rather than relying on labeled examples, the LLM is provided with explicit annotation guidelines that detail the exact definitions of the temporal and causal relations, alongside the strict JSON formatting requirements, directly within the prompt.

In our framework, given a narrative and a fixed set of extracted events, the LLM is prompted to predict semantic relations exclusively between these events according to following predefined relation schemas:

- Temporal-only relations (e.g. BEFORE, OVERLAP),
- Causal-only relations (e.g. CAUSE, ENABLE),
- Joint temporo-causal modeling (e.g. CAUSE_BEFORE).

For each pair of extracted events $(e_i, e_j)$, the model predicts:

$$T(e_i, e_j) \quad \text{and/or} \quad C(e_i, e_j),$$

depending on the experimental condition. This modular setup allows us to evaluate the independent and combined contributions of the temporal and causal structure. As an exemplar, consider the sentence:

"The army attacked the city, causing civilians to flee."

Event extraction yields two events: $e_1$: Attack ("attacked") and $e_2$: Movement ("flee").

The temporal annotation identifies $e_1$ BEFORE $e_2$, while the causal annotation identifies $e_1$ CAUSES $e_2$.

## 3.3 Representing Narratives

After extracting event triggers and annotating temporal and causal relations, each narrative is represented as a structured graph consisting of event nodes and inter-event links. To enable retrieval-based evaluation, these structured representations are linearized into textual sequences and encoded into dense vector embeddings.

Following the representation-level similarity approach of [5], we treat the narrative structure as a transformable textual object that can be embedded using sentence encoders based on large language models [8]. For each narrative, we construct multiple structured views corresponding to our experimental conditions:

- **Event-only**: list of event types (trigger–type pairs),
- **Event + Temporal**: events augmented with temporal relations,
- **Event + Causal**: events augmented with causal relations,
- **Event + Temporal + Causal (independent)**: events with both independently inferred relation sets,
- **Event + Joint Tempero-Causal**: events with jointly modeled semantic links.

Each structured representation is deterministically linearized into a canonical textual format (i.e., an event list ordered by their appearance in the original text, followed by relation triples) to ensure consistent encoding across narratives. Recent evaluations of narrative embeddings indicate that transformer-based encoders exhibit minimal reliance on strict sequencing [7]. We then compute embeddings separately for each representation type. Hence, the design allows controlled ablation of structural components and isolates the (possible) contribution of temporal and causal enrichment to similarity retrieval.

For encoding, we employ the instruction-tuned E5 embedding model [5, 8, 28]. Similarly to [5], each linearized narrative representation is encoded independently in a dense vector of fixed-length. Given a query narrative, similarity to other narratives is computed using cosine similarity in the embedding space. Since high-dimensional representations can introduce noise and suffer

from the curse of dimensionality, we will consider applying a dimensionality reduction method: Singular Value Decomposition (SVD). SVD was shown to effectively reduce variance from superficial lexical differences while preserving deep semantic and structural features [5].

To preserve experimental control, embeddings are generated separately for each structural condition (event-only, event+temporal, event+causal, etc.) rather than concatenating heterogeneous structural components into a single joint representation. With this, we ensure that the observed performance differences reflect only structural enrichment, rather than increased input length, dimensionality, or uncontrolled mixing of different structural components.

## 3.4 Evaluation via Narrative Similarity

*3.4.1 Evaluation Framework and Metrics.* To evaluate the impact of structural enrichment on narrative representation, we adopt a retrieval-based evaluation framework. Narrative similarity is operationalized as the ability of a model to retrieve alternative tellings of the same underlying story from a corpus of candidate narratives. This follows prior work framing narrative similarity as an information retrieval task [3, 8]. All retrieval experiments are performed using cosine similarity over dense embedding vectors. Narratives are ranked in decreasing similarity, and retrieval quality is measured using Precision@1 (P@1), Precision@N (P@N), Mean Average Precision (MAP), Normalized Discounted Cumulative Gain (NDCG) and R-Precision [3, 8, 14].

*3.4.2 Structural Conditions.* Retrieval experiments are conducted separately for each representation type: event-only, event+temporal, event+causal, event+temporal+causal (both semantics extracted independently), and event+temperocausal modeling. This controlled setup allows us to isolate the independent and combined contributions of temporal and causal structure to narrative similarity.

*3.4.3 Lexical and Embedding Baselines.* To contextualize structural gains, we include traditional lexical representation baselines:

- Bag-of-Words (BoW) [3],
- TF-IDF [10],

as well as strong embedding baselines:

- Sentence-BERT [22],
- E5 (instruction-tuned cost-efficient embedding model) [28].

Lexical baselines primarily capture surface overlap and named entity similarity, allowing us to assess the extent to which structural modeling surpasses entity-driven similarity signals. Embedding baselines provide a comparison against dense semantic representations that implicitly encode narrative information without explicit structural constraints.

*3.4.4 Comparison to Narrative-Specific Embeddings.* We compare our structured representations to StoryEmbed [8], using their released pre-trained model. StoryEmbed encodes full narrative summaries and is trained contrastively to retrieve multiple tellings of the same story. Because it operates on full text, implicitly capturing named entities, stylistic cues, and global semantics, it represents a strong upper bound for retrieval performance. In contrast, our representations are derived from structured event graphs. Rather than attempting to outperform StoryEmbed directly, we evaluate whether temporo-causal enrichment narrows the performance gap without sacrificing interpretability and structural grounding.

*3.4.5 Statistical Significance.* To ensure that the incremental improvements resulting from temporal and causal structural enrichment are empirically meaningful and not merely artifacts of dataset variation, we will test the performance differences between models for statistical significance. Because the expected gains between ablations might be relatively small, and because narrative similarity retrieval scores do not typically follow a normal distribution, we will employ non-parametric statistical testing, such as the Mann-Whitney U test, to robustly compare model outputs [8].

*3.4.6 Robustness Across Text Characteristics.* Given the large scale and variance of the Tell Me Again! dataset, aggregate performance metrics may obscure specific strengths or weaknesses of the structural models. To rigorously evaluate robustness, we will additionally test retrieval performance across distinct data subsets partitioned by varying text characteristics. Following recent evaluation methodologies for semantic event extraction [11], we will partition the dataset to isolate different linguistic, syntactic, or semantic phenomena. Analyzing performance differences on these specific subsets will reveal whether the addition of temporal and causal structures provides robust retrieval improvements across diverse narrative contexts.

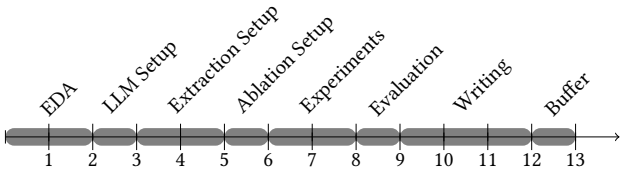
*3.4.7 Error Analysis.* Beyond aggregate metrics, we conduct qualitative error analysis to examine retrieval under different structural conditions. For selected queries, we manually inspect top-ranked results and classify errors into three types: (i) lexical bias, where retrieval is driven by shared entities or surface overlap; (ii) structural alignment, where different vocabulary but similar event progression leads to correct retrieval; and (iii) structural mismatch, where temporal or causal modeling introduces spurious similarity.

# 4 Risk Assessment

First, EDA might reveal some structural limitations in the chosen corpus (Tell me again! [9]), such as insufficient narrative variance or high lexical overlap. If so, alternative strategies (e.g., shifting to cross-dataset evaluation) will be implemented to preserve the validity of the research design. Second, while BERT-based embedding models provide strong baselines, fine-tuning large encoders (if such training is needed) may yield computational and financial constraints. If so, lower-cost alternatives (from the MTEB embedder leaderboard [1]) will be selected instead.

# 5 Project Plan

As shown in the figure below, the project progresses from the validation of the dataset and the feasibility testing of the model through pipeline development, experimentation and evaluation. The last four weeks are reserved for writing and buffer time.



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